

```
=====
## What is Pandas?
=====
```

```
** Pandas is a Python library used for data analysis and data manipulation.
```

```
** Pandas is mainly used to work with structured data.
```

```
** Structured data means data stored in rows and columns.
```

```
### Example :
```

- 1) Student Marks Data
- 2) Employees salary data
- 3) Sales data
- 4) Course Enrollment Data
- 5) CSV Files
- 6) Excel Files

```
=====
## Why Pandas?
=====
```

```
=> Pandas is used for
```

```
** Reading data
```

```
** Writing data
```

```
** Cleaning data
```

```
** Filter data
```

```
** Sorting data
```

```
** Grouping data
```

```
** Analyze data
```

```
** Working with CSV files
```

```
** Working with Excel files
```

```
** Handling missing values
```

```
=====
## Pandas Installation
=====
```

Pandas is an external library.

We need to install it using pip.

Command :

```
pip install pandas
```

```
=====
## Main Data Structures in Pandas
=====
```

**\*\* Pandas has mainly two important data structures:**

1) Series

2) DataFrame

```
=====
## What is Series?
=====
```

=> Series is a one-dimensional data structure in Pandas.

=> It is like a single column

**## Example :**

```
import pandas as pd
```

```
marks = pd.Series([80, 90, 75, 60])
```

```
print(marks)
```

Output :

```
0    80
1    90
2    75
3    60
```

```
dtype: int64
```

```
=====
## What is DataFrame?
=====
```

**\*\* DataFrame is a two-dimensional data structure in Pandas.**

**\*\* It stores data in rows and columns.**

**\*\* It is like an Excel sheet or database table.**

```
---
students = {
    "name": ["Ravi", "Sita", "Kiran"],
    "course": ["Python", "Java", "DevOps"],
    "marks": [80, 90, 75]
}
```

```
df = pd.DataFrame(students)
print(df)
```

```
----
```

```
=====
## Difference Between Series and DataFrame
=====
```

```
** Series ::
```

```
-- One dimensional data
-- Like a single column
-- Created using pd.Series( )
```

```
** DataFrame ::
```

```
-- Two dimensional data
-- Rows and Columns
-- Like Excel sheet or DB table
-- Created using pd.DataFrame ( )
```

```
=====
## Create Series with Custom Index
=====
```

```
# Create series with custom index
```

```
courses = pd.Series(
    ["Python", "JAVA", "DevOps"],
    index = ["C1", "C2", "C3"]
)
```

```
print(courses)
print(courses["C1"])
```

```
=====
## Write DataFrame to CSV File
=====
```

```
** to_csv( ) function is used to write DataFrame data into csv file.
```

```
---
```

```
import pandas as pd
```

```
students = {
    "name": ["Ravi", "Sita", "Kiran"],
    "course": ["Python", "Java", "DevOps"],
    "marks": [80, 90, 75]
}
```

```
df = pd.DataFrame(students)
```

```
df.to_csv("students.csv", index=False)
```

```
print("CSV file created successfully")
```

```
---
```

```
=====
## Reading Data From CSV File
=====
```

```
** CSV means Comma Separated Values.

** Pandas can read CSV files using read_csv().
```

```
---
```

```
import pandas as pd

df = pd.read_csv("students.csv")

print(df)
```

```
---
```

```
=====
Working with Excel
=====
```

```
=> Install "openpyxl" to work with excel files using pandas
```

```
=> To work with excel we have below functions
```

- 1) to\_excel ( )
- 2) read\_excel ( )

```
-----
```

```
import pandas as pd

students = {
    "name": ["Ravi", "Sita", "Kiran"],
    "course": ["Python", "Java", "DevOps"],
    "marks": [80, 90, 75]
}

df = pd.DataFrame(students)
df.to_excel("Students.xlsx", index=False)

print("-----Excel Created Successfully-----")

df = pd.read_excel("Students.xlsx")
print(df)
```

```
-----
```

```
=====
Important functions
=====
```

- 1) head ( ) : displays first 5 rows by default.
- 2) tail ( ) : displays last 5 rows by default.
- 3) info ( ) : displays DataFrame information

```
-- number of rows
-- number of columns
-- column names
-- data types
-- non-null values
```

4) describe ( ) : gives statistical summary for numeric columns.

```
-----

import pandas as pd

employees = {
    "EmpId": [101, 102, 103, 104, 105, 106],
    "Name": ["Ravi", "Sita", "Kiran", "Rahul", "Priya", "Anil"],
    "Department": ["IT", "HR", "IT", "Sales", "HR", "IT"],
    "Salary": [50000, 45000, 70000, 40000, 48000, 65000],
    "Experience": [2, 5, 8, 1, 4, 7]
}

df = pd.DataFrame(employees)
print(df)
print("-----")
print(df.head(3))
print("-----")
print(df.tail(3))
print("-----")
print(df["Name"])
print("-----")
print(df[["Name", "Salary"]])

print("-----")

print(df[df["Salary"] > 50000])

print("-----")

print(df[df["Department"] == "IT"])

print("-----")

print(df[(df["Department"] == "IT") & (df["Salary"] > 60000)])

print("-----")

print(df[(df["Department"] == "IT") | (df["Salary"] > 60000)])

print("-----")

print(df.sort_values("Salary"))

print(df.sort_values("Salary", ascending=False))

print(df.sort_values(["Department", "Salary"]))

print("-----")

df["Bonus"] = df["Salary"] * 0.10

print(df)
```

```
print("-----")

df["Total Salary"] = df["Salary"] + df["Bonus"]

print(df)

print("-----")

df.drop(columns=["Bonus"], inplace=True)
print(df)

print("-----")

print(df.groupby("Department")["Salary"].mean())
print(df.groupby("Department")["Salary"].max())
print(df.groupby("Department")["Salary"].min())
print(df.groupby("Department")["Salary"].sum())

print("-----")

print(df["Department"].unique())
print(df["Department"].nunique())

print("-----")

emps = {
    "Name":["Ravi","Sita","Kiran","Rahul"],
    "Salary":[50000,None,65000,None]
}

ndf = pd.DataFrame(emps)

print(ndf)
#print(df.isnull())
print("-----")

ndf["Salary"] = ndf["Salary"].fillna(0)
print(ndf)

-----

=====Assignment-1=====

students = {
    "Name":["Ravi","Sita","Kiran","Rahul","Priya"],
    "Course":["Python","Java","Python","DevOps","Python"],
    "Marks":[85,90,75,60,95]
}

df = pd.DataFrame(students)

1) Show only python students
2) Show students with marks > 80
3) Sort students based on marks
4) Find Highest marks
5) Find Avg Marks
6) Count students in each course
7) Add Grade Column
8) Save the Student data into CSV
```

- 9) Read the CSV file
- 10) Store into Excel

=====Assignment-2=====

```
products = {  
    "Product":["Laptop","Mouse","Keyboard","Monitor","Printer"],  
    "Category":["Electronics","Accessories","Accessories","Electronics","Electronics"],  
    "Price":[60000,500,1200,15000,10000],  
    "Stock":[15,200,120,25,18]  
}  
  
df = pd.DataFrame(products)
```

- 1) Show products above â,¹10,000.
- 2) Sort by price.
- 3) Count products by category.
- 4) Find average price by category.
- 5) Add GST (18%).
- 6) Calculate Final Price.
- 7) Save to CSV.
- 8) Read CSV.